

MATH1061/7861 Glossary

Basic Terms

Integer A number with no fractional part. It includes the counting numbers $\{1, 2, 3, \dots\}$, zero $\{0\}$, and the negatives of the counting numbers $\{-1, -2, -3, \dots\}$. The set of all integers is written as $\mathbb{Z}$.

Even Number An integer that can be written in the form $2k$ for some integer k . Every integer is either odd or even (but not both). The last digit of an even number must be 0, 2, 4, 6 or 8. Some examples of even numbers are -12 , 0, 2, 8 and 42. Note that 0 is even, as $0 = 2 \times 0$. The set of all even integers is sometimes written as $\mathbb{Z}_{\text{even}}$.

Odd Number An integer that can be written in the form $2k + 1$ for some integer k . Every integer is either odd or even (but not both). The last digit of an odd number must be 1, 3, 5, 7 or 9. Some examples of odd numbers are -3 , 1, 7 and 35. The set of all odd integers is sometimes written as $\mathbb{Z}_{\text{odd}}$.

Rational Number A number that can be expressed in the form $\frac{a}{b}$ for some integers a, b with $b \neq 0$. The set of all rational numbers is written as $\mathbb{Q}$.

Real Number Any point on the number line, such as 1, 15.82, -0.1 , $3/4$, $\pi/2$, etc. Positive or negative, large or small, whole numbers or decimal numbers are all real numbers. The set of all real numbers is written as $\mathbb{R}$.

Irrational Number A real number that cannot be expressed in the form $\frac{a}{b}$ for some integers a, b with $b \neq 0$. Every real number is either rational or irrational (but not both). The set of all irrational numbers is often written as $\mathbb{R} \setminus \mathbb{Q}$ or $\overline{\mathbb{Q}}$.

Prime Number An integer greater than 1 whose only positive divisors are 1 and itself. The first few prime numbers are 2, 3, 5, 7, 11 and 13.

Composite Number An integer n greater than 1 such that $n = rs$ for some positive integers r and s with $r \neq 1$ and $s \neq 1$. A composite number is always the product of 2 or more prime numbers. Note that 0 and 1 are neither prime nor composite.

Factor/Divisor An integer that divides another integer with no remainder. If k is a factor of n , then there is some integer a such that $ak = n$. We write $k \mid n$ to indicate that k divides n .

Summation The addition of a sequence of numbers, often written using the symbol $\sum$. For example,

$$\sum_{i=3}^5 i^2 = 3^2 + 4^2 + 5^2 = 50.$$

Factorial The factorial of a positive integer n is the product of all the positive integers less than or equal to n . It is written as $n!$. For example $5! = 5 \times 4 \times 3 \times 2 \times 1 = 120$.

Coefficient A constant by which a variable is multiplied. For example, the coefficient of x^2 in $3x^3 - 2x^2 + 5$ is -2 . (Refer to Video 40: Binomial Coefficients)

Logical Equivalence

Statement/Proposition A declarative sentence with a definite meaning, with a truth value that's either true (T) or false (F). It can never be both, neither, or somewhere in between. (Refer to Video 1: Logical Form)

Negation The negation of a statement A is written as $\sim A$ or $\neg A$ and is often called “not A ”. If A is true, then its negation is false; if A is false, then its negation is true. (Refer to Video 1: Logical Form)

Conjunction The conjunction of two statements is true if and only if both statements are true. The conjunction of two statements P and Q is written as $P \wedge Q$ (whenever you see $\wedge$ read “and”). (Refer to Video 1: Logical Form)

Disjunction The disjunction of two statements is true if and only if at least one of the statements is true. The disjunction of two statements P and Q is written as $P \vee Q$ (whenever you see $\vee$ read “or”). (Refer to Video 1: Logical Form)

Logical Equivalence Two statement forms are logically equivalent if they have the same truth value for any assignment of truth values to its variables. If two statement forms P and Q are logically equivalent, we write $P \equiv Q$. (Refer to Video 2: Logical Equivalence)

Tautology A statement form which is always true, regardless of the truth values of its variables. (Refer to Video 2: Logical Equivalence)

Contradiction A statement form which is always false, regardless of the truth values of its variables. (Refer to Video 2: Logical Equivalence)

Conditional statement A statement $P \rightarrow Q$, read as “ P implies Q ” or “if P then Q ”. It is logically equivalent to $\sim P \vee Q$. The first statement (P in this case) is sometimes called the hypothesis, and the second statement (Q in this case) is sometimes called the conclusion. (Refer to Video 3: Conditional Statement)

Biconditional statement A statement $P \leftrightarrow Q$, which is read as “ P if and only if Q ”. It is logically equivalent to $(P \rightarrow Q) \wedge (Q \rightarrow P)$. (Refer to Video 3: Conditional Statement)

Contrapositive The contrapositive of the conditional statement $P \rightarrow Q$ is the statement $\sim Q \rightarrow \sim P$. It is logically equivalent to the original statement. (Refer to Video 3: Conditional Statement)

Valid and Invalid Arguments

Valid An argument is valid if, whenever all of the premises are true, the conclusion is also true. (Refer to Video 4: Valid and invalid arguments)

Invalid An argument is invalid if it is possible to have a situation in which all of the premises are true but the conclusion is false. (Refer to Video 4: Valid and invalid arguments)

Quantified Statements

Predicate A sentence that contains finitely many variables, and becomes a statement when the variables are given specific values. (Refer to Video 6: Quantified statements)

Truth Set The truth set of a predicate $P(x)$ is the set of all values in the domain that, when assigned to x , make $P(x)$ a true statement. (Refer to Video 6: Quantified statements)

Quantifiers The universal quantifier “for all”, written as $\forall$, is used to say that a predicate is true for every member of a domain. The existential quantifier “there exists”, written as $\exists$, is used to say that a predicate is true for at least one member of the domain. (Refer to Video 6: Quantified statements)

Divisibility & Modular Arithmetic

Factorisation A method of writing numbers as the product of their factors.

Unique Prime Factorisation Theorem The theorem stating that every integer greater than 1 either is a prime number or can be represented as a product of prime numbers. This representation is unique up to the ordering of the primes. For example, $1200 = 2^4 \times 3 \times 5^2$ (Refer to Video 13: Divisibility)

Floor Given any $x \in \mathbb{R}$, the floor of x , denoted $\lfloor x \rfloor$, is the greatest integer less than or equal to x . For example, $\lfloor 2.3 \rfloor = 2$, $\lfloor 8 \rfloor = 8$ and $\lfloor -1.5 \rfloor = -2$. (Refer to Video 14: Modular Arithmetic)

Ceiling Given any $x \in \mathbb{R}$, the ceiling of x , denoted $\lceil x \rceil$, is the smallest integer greater than or equal to x . For example, $\lceil 2.3 \rceil = 3$, $\lceil 8 \rceil = 8$ and $\lceil -1.5 \rceil = -1$. (Refer to Video 14: Modular Arithmetic)

Remainder The number left over when one integer is divided by another in integer division. For example, 16 divided by 3 has a remainder of 1, as $16 = 5 \times 3 + 1$. If r is the remainder of n divided by d , then $0 \leq r < d$. (Refer to Video 14: Modular Arithmetic)

Modulo Integers a and b are considered equivalent modulo n if n divides $a - b$. We sometimes also say that “ a is congruent to b modulo n ” and write $a \equiv b \pmod{n}$. (Refer to Video 14: Modular Arithmetic)

Euclidean Algorithm An algorithm for finding the greatest common divisor of two positive integers. (Refer to Video 15: The Euclidean Algorithm)

Sequences

Sequence An ordered list of elements. Each element is called a term. (Refer to Video 16: Sequences)

Explicit formula A formula that provides a way to calculate the n^{th} term of a sequence without the previous terms in the sequence needing to be calculated. (Refer to Video 16: Sequences)

Recursive definition A definition given by means of recursion. The definition is composed of two parts: the base case, which is an explicit definition of a few terms, and the recursive formula, which is the rule used to calculate new terms based on previous terms. (Refer to Video 19: Recursive Definitions)

Set Theory

Set An unordered collection of elements. (Refer to Video 21: Set theory definitions)

Empty Set The set with no elements, written $\{\}$ or $\emptyset$. (Refer to Video 21: Set Theory Definitions)

Set Equality Two sets are equal if they have the same elements. Note that order and duplicates don't matter, so $\{1, 2\} = \{2, 1, 2\}$. (Refer to Video 21: Set Theory Definitions)

Subset A set Y is a subset of X (written $Y \subseteq X$) if every element in Y also occurs in X . For example, $\{1, 2, 3\} \subseteq \{1, 2, 3, 4, 5\}$. (Refer to Video 21: Set Theory Definitions)

Proper subset A set Y is a proper subset of X (written $Y \subset X$ or $Y \subsetneq X$) if Y is a subset of X , but Y is not equal to X . (Refer to Video 21: Set Theory Definitions)

Union The union of two sets X and Y (written $X \cup Y$) is the set containing all elements of X and all elements of Y . (Refer to Video 21: Set theory definitions)

Intersection The intersection of two sets X and Y (written $X \cap Y$) is the set containing all elements that occur in both X and Y . (Refer to Video 21: Set Theory Definitions)

Power Set The power set of a set A (written $\mathcal{P}(A)$) is the set of all possible subsets of A . For example, $\mathcal{P}(\{1, 2\}) = \{\emptyset, \{1\}, \{2\}, \{1, 2\}\}$. (Refer to Video 22: More Definitions and Examples of Sets)

Cartesian product For sets A and B , the Cartesian product $A \times B$ is the set consisting of all ordered pairs (a, b) where $a \in A$ and $b \in B$. For example, $\{1, 2\} \times \{3, 4\} = \{(1, 3), (1, 4), (2, 3), (2, 4)\}$. (Refer to Video 22: More Definitions and Examples of Sets)

Functions

One-to-One/Injective A function $f : X \rightarrow Y$ is one-to-one if for any $a, b \in X$ with $a \neq b$ we have $f(a) \neq f(b)$. This is equivalent to saying that if $f(a) = f(b)$ for some $a, b \in X$, then $a = b$. (Refer to Video 25: One-one, Onto and Inverse Functions)

Onto A function $f : X \rightarrow Y$ is onto if for any $y \in Y$ there exists some $x \in X$ such that $f(x) = y$. A function is onto if and only if the image of the function is equal to its codomain. (Refer to Video 25: One-one, Onto and Inverse Functions)

Inverse An inverse function is a function that undoes the action of the other function. A function g is the inverse of an injective function f if whenever $y = f(x)$ then $x = g(y)$. We often write f^{-1} to denote the inverse of f . (Refer to Video 25: One-one, Onto and Inverse Functions)

Bijective A bijective function $f : X \rightarrow Y$ is a function that is both one-to-one and onto. Such a function is called a bijection. (Refer to Video 25: One-one, Onto and Inverse Functions)

Composition For any functions $g : A \rightarrow B$ and $f : B \rightarrow C$, the composition of f and g , written $f \circ g$, is the function $f \circ g : A \rightarrow C$ defined by $(f \circ g)(a) = f(g(a))$, for all $a \in A$. (Refer to Video 26: Composition of Functions)

Cardinality

Cardinality A measure of the size of a set. The cardinality of a set A is written as $|A|$. For finite sets, the cardinality is the number of elements in A . For example, $|\{35, 42\}| = 2$. (Refer to Video 27: Cardinality)

Countable set A set with the same cardinality as some subset of $\mathbb{Z}_{\geq 0}$. A countable set can have finite many or infinitely many elements. (Refer to Video 28: Countable and Uncountable Sets)

Uncountable set An infinite set that contains too many elements to be countable. A set A is uncountable if there is no bijection from A to $\mathbb{Z}$. (Refer to Video 28: Countable and Uncountable Sets)

Set Relations

Set Relation A binary relation R between two sets A and B is a subset of the Cartesian product $A \times B$. If $(a, b) \in R$, we say that a and b are related under R and write aRb . (Refer to Video 30: Reflexivity, Symmetry, Transitivity)

Reflexive A binary relation R over a set A is reflexive if every element of A is related to itself. This can be written as $\forall a \in A, aRa$. (Refer to Video 30: Reflexivity, Symmetry, Transitivity)

Symmetric A binary relation R over a set A is symmetric if aRb implies that bRa for all $a, b \in A$. (Refer to Video 30: Reflexivity, symmetry, transitivity)

Transitive A binary relation R over a set A is transitive if, for any $a, b, c \in A$ such that aRb and bRc , we have aRc . (Refer to Video 30: Reflexivity, Symmetry, Transitivity)

Antisymmetric A relation R over a set A is called antisymmetric if for any $a, b \in A$, aRb and bRa imply that $a = b$. Note that a relation being antisymmetric is different to the relation being not symmetric. (Refer to Video 32: Partial order relations)

Equivalence Relation A binary relation that is reflexive, symmetric, and transitive. (Refer to Video 31: Equivalence Relations)

Partition A partition of a set is a grouping of the set's elements into non-empty subsets, in such a way that every element is included in one and only one of the subsets. The union of all the parts of a partition is equivalent to the whole set, and the intersection of any two distinct parts of a partition is the empty set. (Refer to Video 31: Equivalence Relations)

Partial Order Relation A binary relation R that is reflexive, antisymmetric, and transitive. (Refer to Video 32: Partial order relations)

Total Order Relation A partial order relation R on a set A such that for all $a, b \in A$, either aRb or bRa (or both). (Refer to Video 30: Reflexivity, Symmetry, Transitivity)

Groups

Group A set G of elements together with a binary operation $*$ (called the group operation) satisfying four properties: closure, associativity, the identity property, and the inverse property. A group is commonly written as $(G, *)$. (Refer to Video 33: Definitions and Examples of Groups)

Subgroup If $(G, *)$ and $(H, *)$ are groups and $H \subseteq G$, then $(H, *)$ is a subgroup of $(G, *)$. A subset of G is a group if it is closed, contains the identity and contains inverses. (Refer to Video 34: Elementary Properties of Groups)

Field A field $(F, +, \cdot)$ is a set F with two binary operations $+$ and $\cdot$ such that $(F, +)$ is an abelian group; $(F - \{0\}, \cdot)$ is an abelian group where 0 is the identity of $(F, +)$; and for any $a, b, c \in F$ we have $a \cdot (b + c) = a \cdot b + a \cdot c$. (Refer to Video 36: Definitions and Examples of Fields)

Counting

Probability A number between 0 (impossible) and 1 (certain) expressing the likelihood of an event occurring. For instance, the sun will rise tomorrow has a probability of 1. It can be expressed as a fraction, a decimal, or a percentage. The probability of an event X occurring is written $\mathbb{P}(X)$. (Refer to Video 39: Introduction to probability)

Pigeonhole Principle The pigeonhole principle states that if $n + 1$ objects are placed into n boxes, then there is at least one box containing two objects. (Refer to Video 42: The Pigeonhole Principle)

Graph Theory

Graph A graph is a pair (V, E) composed of two sets: the set V of vertices, and the set E of edges, which consists of unordered pairs of vertices. (Refer to video 43 Introduction to Graph Theory)

Simple graph A simple graph is a graph that contains no loops and no multiple edges. (Refer to video 43 Introduction to Graph Theory)

Adjacent Vertices Two vertices u and v of a graph G are adjacent if there is an edge between u and v in G . (Refer to Video 44: Walks, trails and circuits)

Degree The degree of a vertex v (written $\deg(v)$) is the number of edges incident to v .

Walk A walk is a sequence $v_0, e_1, v_1, e_2, v_2, \dots, v_n$, where each v_i is a vertex, and each e_i is an edge with endpoints v_{i-1} and v_i . (Refer to Video 44: Walks, Trails and Circuits)

Cycle A walk that starts and ends at the same vertex but otherwise has no repeated vertices. (Refer to Video 44: Walks, trails and circuits)

Non-Trivial Cycle A cycle consisting of more than one vertex (Refer to Video 44: Walks, trails and circuits)

Trail A trail is a walk where all the edges are distinct. (Refer to Video 44: Walks, Trails and Circuits)

Circuit A circuit is a trail that starts and ends at the same vertex. (Refer to Video 44: Walks, Trails and Circuits)

Acyclic A graph that does not contain any non-trivial cycles.

Connected Graph A graph such that for any pair of vertices, there is a walk between them. (Refer to Video 44: Walks, Trails and Circuits)

Eulerian Trail A trail that uses every edge of the graph exactly once. (Refer to Video 44: Walks, Trails and Circuits)

Eulerian Circuit A circuit that uses every edge of the graph exactly once. (Refer to Video 44: Walks, Trails and Circuits)

Regular graph A graph in which all vertices have the same degree. (Refer to Video 44: Walks, Trails and Circuits)

Cycle Graph A graph consisting of n vertices $v_1, \dots, v_n$ and n edges given by the set $\{\{v_1, v_2\}, \dots, \{v_{n-1}, v_n\}, \{v_n, v_1\}\}$. It is denoted by C_n . (Refer to Video 44: Walks, Trails and Circuits)

Adjacency Matrix The adjacency matrix for a graph with vertices $v_1, v_2, \dots, v_n$ is the $n \times n$ matrix $A = [a_{ij}]$ where each entry a_{ij} is the number of edges with endpoints v_i, v_j . (Refer to Video 45: Matrix Representations of Graphs)

Incidence Matrix The incidence matrix for a graph with vertices $v_1, v_2, \dots, v_n$ and edges $e_1, e_2, \dots, e_m$ is the $n \times m$ matrix $A = [a_{ij}]$ where each entry a_{ij} is the number of times vertex v_i is incident with edge e_j . (Refer to Video 45: Matrix Representations of Graphs)

Tree An acyclic, connected, undirected graph. (Refer to Video 46: Trees)